**NIUS 21(2025)-PHYSICS**
**INTERIM REPORT**

**Submitted by:**
Gopika Gireesh
Integrated MSc Mathematics
Cochin University of Science and Technology
Batch: 21(2025)
Project Code: T8

**Mentor:**
Prof. Anuradha Misra, F.N.A.Sc
Emeritus Scientist CSIR
UMDAE-CEBS
University of Mumbai

# 1 Introduction

The present project was initially undertaken with the objective of developing a strong theoretical foundation in elementary particle physics and relativistic quantum mechanics. The study began with a systematic review of the structure of elementary particles and their classification within the framework of the Standard Model, fundamental interactions and their representation through lowest-order Feynman diagrams, enabling a diagrammatic understanding of particle scattering processes. The project later progressed to relativistic quantum mechanics where the formalism of four-vectors along with Lorentz invariance and conservation of four-momentum were studied in detail. Problems involving particle collisions were analyzed in both the center-of-mass and laboratory frames, strengthening conceptual and computational understanding of relativistic kinematics. The derivation and application of scattering cross-section formulae were also examined, along with an introduction to perturbation theory as a tool for calculating transition amplitudes.

Subsequently, relativistic wave equations were explored. The Klein–Gordon equation was studied as a scalar relativistic equation, followed by a detailed examination of the Dirac equation, including its Lorentz-covariant structure. Work on Dirac spinors and spin-dependent contributions to scattering processes is currently in progress. An introductory study of quantum field theory was also initiated through the formulation and quantization of the Klein–Gordon field, highlighting the conceptual shift from single-particle wavefunctions to operator-valued fields.

However, the research direction has currently been refined towards Quantum Information Theory, motivated by its foundational connection to quantum mechanics and its growing significance in modern theoretical and applied physics. The current phase of the project focuses on establishing the mathematical and conceptual framework of quantum information theory. This report summarizes the progress achieved so far and outlines the planned direction for the subsequent phase of work.

# 2 Foundations of Particle Physics

## 2.1 The Standard Model

The Standard Model (SM) of particle physics describes the fundamental constituents of matter and their interactions via the electromagnetic, weak, and strong forces. The theory

classifies elementary particles into two broad categories: *fermions* (matter particles) and *bosons* (force carriers).

### Matter Particles: Fermions

Fermions are spin-$\frac{1}{2}$ particles and are arranged into three generations. Each generation contains two quarks and two leptons. Important conserved quantum numbers include charge (Q), baryon number (B), Lepton number (muon number, tau number, electron number), upness (U), downess (D), strangeness(S) , charm(C), topness(T), bottomness(B).Quarks carry color charge and participate in strong, weak, and electromagnetic interactions.Leptons do not carry color charge and therefore do not participate in strong interactions.

| **Generation** | **Up-type** | **Down-type** |
|---|---|---|
| 1 | $u$ | $d$ |
| 2 | $c$ | $s$ |
| 3 | $t$ | $b$ |

| **Generation** | **Charged Lepton** | **Neutrino** |
|---|---|---|
| 1 | $e^-$ | $\nu_e$ |
| 2 | $\mu^-$ | $\nu_\mu$ |
| 3 | $\tau^-$ | $\nu_\tau$ |

### Force Carriers: Gauge Bosons

Each fundamental interaction is mediated by boson(s):

| **Interaction** | **Mediator** |
|---|---|
| $Strong$ | Gluons ($g$) |
| $Electromagnetic$ | $Photons(\gamma)$ |
| $Weak$ | $W^+, W^-, Z^0$ |

### The Quark Model and Flavor Quantum Numbers

Hadrons are composite particles made of quarks. They are classified into Baryons($qqq$) and Mesons($q\bar{q}$).

**Baryon Decuplet (qqq States)**

| Particle | Quark Content | $B$ | $S$ | $Q$ |
|---|---|---|---|---|
| $\Delta^{++}$ | $uuu$ | 1 | 0 | $+2$ |
| $\Delta^{+}$ | $uud$ | 1 | 0 | $+1$ |
| $\Delta^{0}$ | $udd$ | 1 | 0 | 0 |
| $\Delta^{-}$ | $ddd$ | 1 | 0 | $-1$ |
| $\Sigma^{*+}$ | $uus$ | 1 | $-1$ | $+1$ |
| $\Sigma^{*0}$ | $uds$ | 1 | $-1$ | 0 |
| $\Sigma^{*-}$ | $dds$ | 1 | $-1$ | $-1$ |
| $\Xi^{*0}$ | $uss$ | 1 | $-2$ | 0 |
| $\Xi^{*-}$ | $dss$ | 1 | $-2$ | $-1$ |
| $\Omega^{-}$ | $sss$ | 1 | $-3$ | $-1$ |

**Meson Nonet (q$\bar{q}$ States)**

| Meson | Quark Content | $B$ | $S$ | $Q$ |
|---|---|---|---|---|
| $\pi^{+}$ | $u\bar{d}$ | 0 | 0 | $+1$ |
| $\pi^{0}$ | $\frac{1}{\sqrt{2}}(u\bar{u} - d\bar{d})$ | 0 | 0 | 0 |
| $\pi^{-}$ | $d\bar{u}$ | 0 | 0 | $-1$ |
| $K^{+}$ | $u\bar{s}$ | 0 | $+1$ | $+1$ |
| $K^{0}$ | $d\bar{s}$ | 0 | $+1$ | 0 |
| $\bar{K}^{0}$ | $s\bar{d}$ | 0 | $-1$ | 0 |
| $K^{-}$ | $s\bar{u}$ | 0 | $-1$ | $-1$ |
| $\eta, \eta'$ | mixing states | 0 | 0 | 0 |

**Role of the Higgs Boson**

The Higgs field has a non–zero vacuum expectation value and allows particles to obtain mass through their interaction with this field. Without the Higgs mechanism, the $W^{\pm}$ and $Z^{0}$ bosons would be massless, and fermions would not acquire mass. The discovery of the Higgs boson in 2012 confirmed this mechanism.

Despite its experimental success, the Standard Model is incomplete since it does not include gravity, cannot explain dark matter or dark energy, does not explain neutrino masses naturally anddoes not explain matter-antimatter asymmetry fully.These limitations motivate extensions such as Grand Unified Theories,Supersymmetry, and other Beyond Standard Model (BSM) frameworks.

## 2.2 Feynman Diagrams and Interaction Vertices

Feynman diagrams are pictorial representations of particle interactions. Each vertex in the diagram represents a fundamental interaction and is associated with a coupling constant that determines the strength of that interaction.

**(i) Electromagnetic Interaction: QED Vertex**

$$e^- \longleftrightarrow e^- + \gamma$$

The vertex represents the emission or absorption of a photon by a charged fermion. Charge $Q$, lepton number $L$, and four-momentum are conserved at each vertex. The strength of the interaction is governed by the fine structure constant,

$$\alpha = \frac{e^2}{4\pi} \approx \frac{1}{137}.$$

**(ii) Strong Interaction: QCD Vertices**

**a) Quark–Quark–Gluon Vertex:** This vertex preserves Color charge, Baryon number, Electric charge, Flavor quantum numbers (in strong processes).

$$q \longleftrightarrow q + g$$

**b) Gluon Self-Interaction:** Unlike photons, gluons carry color charge. Therefore, QCD predicts:

$$g \longleftrightarrow g + g \qquad g + g \longleftrightarrow g + g$$

**(iii) Weak Interactions**

Weak interactions are mediated by $W^\pm$ and $Z^0$ bosons.

**a) Lepton–Neutrino–$W^-$ vertex:** This vertex changes electric charge and flavor while conserving total lepton number.

$$e^- \longleftrightarrow \nu_e + W^-$$

**b) Neutral Current; Lepton–Lepton–$Z^0$ vertex:** No electric charge is exchanged, and flavor is conserved.

$$e^- \longleftrightarrow e^- + Z^0$$

**c) Quark–Quark–$W^-$ vertex:** Weak interactions allow flavor change:

$$u \longleftrightarrow d + W^+$$

More generally,

$$u \longleftrightarrow d', \quad d' = d\cos\theta_C + s\sin\theta_C.$$

**Cabibbo Mixing:** For the first two generations,

$$\begin{pmatrix} d' \\ s' \end{pmatrix} = \begin{pmatrix} \cos\theta_C & \sin\theta_C \\ -\sin\theta_C & \cos\theta_C \end{pmatrix} \begin{pmatrix} d \\ s \end{pmatrix}.$$

Here $\theta_C$ is the Cabibbo angle ($\approx 13°$). For example, beta decay involves

$$u \to d' + W^+,$$

meaning strange quark contributions are suppressed by $\sin\theta_C$. This explains relative decay probabilities in weak processes.

**Gauge Boson Self-Interactions:** Like gluons, the intermediate vector bosons ($W^\pm$ and $Z^0$) can also couple to themselves and so does some gauge bosons.

$$W^- \longleftrightarrow W^- + Z^0 \qquad Z^0 \longleftrightarrow W^+ + W^-$$

$$W^+ + W^- \longleftrightarrow Z^0 \qquad W^+ + W^- \longleftrightarrow \gamma$$

**Coupling Constants and Their Hierarchy:**

Each interaction is characterized by a coupling constant:

$$\alpha \sim 10^{-2} \quad \text{(EM)},$$
$$\alpha_s \sim 1 \text{ at low energies} \quad \text{(Strong)},$$
$$G_F \sim 10^{-5}\,\text{GeV}^{-2} \quad \text{(Weak)}.$$

Thus, at low energies:

$$\text{Strong} > \text{Electromagnetic} > \text{Weak}.$$

This hierarchy of coupling strengths explains why, in electromagnetic and weak processes, lowest–order (tree-level) Feynman diagrams are often sufficient to obtain accurate predictions. Since the expansion parameter is small ($\alpha \ll 1$), higher–order corrections are typically suppressed.

In contrast, for strong interactions at low energies where $\alpha_s \sim 1$, perturbation theory breaks down and higher–order contributions become significant. This is why perturbative methods are reliable mainly at high energies in QCD, where asymptotic freedom makes $\alpha_s$ small.

**Conservation Laws at Interaction Vertices**

At each vertex, certain conservation laws must be satisfied:

- **Energy and momentum** (always conserved)
- **Electric charge** $Q$ (strictly conserved)

- **Baryon number** $B$ (conserved in SM)
- **Lepton number** $L$ (approximately conserved)
- **Color charge** (strictly conserved in QCD)
- **Strangeness** $S$ (conserved in strong, violated in weak)

Some conservation laws are exact within the Standard Model (e.g., electric charge, color), while others such as strangeness are violated in weak interactions.

Feynman vertices therefore encode both the symmetry structure and dynamical content of the Standard Model. By associating coupling constants and conservation laws to each vertex, one can systematically construct scattering amplitudes and compute measurable cross sections.

## 2.3 Four–Vector Formalism

Relativistic dynamics is most naturally formulated using four–vectors, which ensure manifest Lorentz covariance of physical laws.

### Definition of a Four–Vector

A four–vector is an object that transforms under Lorentz transformations as

$$x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu},$$

where $\Lambda^{\mu}{}_{\nu}$ satisfies $\Lambda^{\mu}{}_{\alpha}\Lambda^{\nu}{}_{\beta} g_{\mu\nu} = g_{\alpha\beta}$.

A general contravariant four–vector is written as $A^{\mu} = (A^0, \mathbf{A})$. Its covariant form is obtained using the Minkowski metric

$$g_{\mu\nu} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix},$$

such that $A_{\mu} = g_{\mu\nu} A^{\nu} = (A^0, -\mathbf{A})$.

A Lorentz boost along the $x$–direction is

$$\Lambda = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

with $\beta = v$ and $\gamma = (1 - \beta^2)^{-1/2}$, under which $x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}$.

The Lorentz–invariant scalar product is

$$A \cdot B = g_{\mu\nu} A^\mu B^\nu = A^0 B^0 - \mathbf{A} \cdot \mathbf{B},$$

so that

$$A^\mu A_\mu = (A^0)^2 - \mathbf{A}^2.$$

Depending on the sign of this invariant quantity, four–vectors are classified as:

$a$)**Timelike** $(A^\mu A_\mu > 0)$
$b$)**Spacelike** $(A^\mu A_\mu < 0)$
$c$)**Lightlike** $(A^\mu A_\mu = 0$

For example, the four–momentum satisfies $p^\mu p_\mu = m^2$.

Thus, massive particles correspond to timelike four–vectors, while photons satisfy $p^\mu p_\mu = 0$, and are therefore lightlike.

### Four–Velocity and Four–Momentum

The spacetime position four–vector is $x^\mu = (t, \mathbf{x})$.

The four–velocity is defined as

$$u^\mu = \frac{dx^\mu}{d\tau} = \gamma(1, \mathbf{v}),$$

where $\tau$ is proper time and $\gamma = (1 - v^2)^{-1/2}$.

The four–momentum is

$$p^\mu = m u^\mu = (\gamma m, \gamma m \mathbf{v}) = (E, \mathbf{p}),$$

with $E^2 = \mathbf{p}^2 + m^2$.

The invariant mass relation follows directly from $p^\mu p_\mu = m^2$.

### Conservation of Four–Momentum

In any relativistic interaction,

$$\sum p^\mu_{\text{initial}} = \sum p^\mu_{\text{final}}.$$

This single equation encodes conservation of both energy and three–momentum.

### Example: Two–Body Scattering

Consider elastic scattering $a + b \to a + b$, with four–momenta $p_1, p_2 \to p_3, p_4$.

Four–momentum conservation gives $p_1 + p_2 = p_3 + p_4$.

Define the Mandelstam invariants

$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_1 - p_4)^2.$$

Because this is a dot product of four–vectors, $s$ is Lorentz invariant.

**In the CM frame**: $\mathbf{p}_1 = -\mathbf{p}_2$.

Thus, $p_1^\mu = (E_1, \mathbf{p}), \quad p_2^\mu = (E_2, -\mathbf{p})$. Then, $s = (E_1 + E_2)^2$.

For identical masses $m$, $E_1 = E_2 = \sqrt{p^2 + m^2}$,

so $s = 4(p^2 + m^2)$.

**In the lab frame**: particle $b$ is initially at rest: $p_2^\mu = (m, 0)$.

Then, $s = (p_1 + p_2)^2 = p_1^2 + p_2^2 + 2p_1 \cdot p_2$.

Since $p_1^2 = m^2$ and $p_2^2 = m^2$, $s = 2m^2 + 2p_1 \cdot p_2$.

Now, $p_1 \cdot p_2 = E_1 m$,

so$s = 2m^2 + 2mE_1$.

**Lorentz Invariance Demonstration:** Equating the two expressions for $s$,

$$4(p^2 + m^2) = 2m^2 + 2mE_1.$$

This relation connects CM momentum with lab energy. The equality holds because $s$ is Lorentz invariant. This example demonstrates two essential features:

- Four–momentum conservation unifies energy and momentum conservation.
- The invariant dot product allows scattering processes to be analyzed consistently across different inertial frames.

## 2.4 Transition Amplitude and Differential Cross Section

To describe scattering processes, we begin by introducing an interaction term into the relativistic Schrödinger equation. For a spinless particle,

$$i\frac{\partial}{\partial t}\psi = (H_0 + V)\psi,$$

where $H_0$ is the free Hamiltonian and $V$ is the interaction potential.

### Transition Amplitude

The transition amplitude from an initial state $|i\rangle$ to a final state $|f\rangle$ is defined as

$$T_{fi} = \langle f|V|i\rangle.$$

In relativistic scattering theory, this matrix element can be written in terms of an invariant amplitude $\mathcal{M}$ as

$$T_{fi} = (2\pi)^4 \delta^{(4)}(p_f - p_i)\,(-i\mathcal{M}).$$

**Electron–Muon Scattering**

Consider spinless electron–muon scattering:

$$e^- + \mu^- \to e^- + \mu^-.$$

The interaction is mediated by photon exchange. We define the currents

$$j_1^\mu = e\,(p_1 + p_3)^\mu, \qquad j_2^\mu = e\,(p_2 + p_4)^\mu,$$

and the electromagnetic field $A^\mu$.

The interaction potential takes the form

$$V \sim j_1^\mu A_\mu.$$

Substituting into the matrix element generates the invariant amplitude $\mathcal{M}$ corresponding to the single photon exchange Feynman diagram. (We do not substitute its explicit expression here, but it is obtained directly from Feynman rules.)

**Transition Rate**

The transition probability per unit spacetime volume is

$$W_{fi} = \frac{|T_{fi}|^2}{VT}.$$

Substituting the expression for $T_{fi}$ yields

$$W_{fi} = (2\pi)^4 \delta^{(4)}(p_f - p_i)\,|\mathcal{M}|^2.$$

**Definition of Cross Section**

The differential cross section is defined as

$$d\sigma = \frac{W_{fi}\,(\text{number of final states})}{\text{initial flux}}.$$

To evaluate this expression, we determine:

**1. Initial Flux** For a general process

$$A(p_1) + B(p_2) \to C(p_3) + D(p_4),$$

the invariant flux factor is

$$F = 4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}.$$

In the center of mass (CM) frame,

$$F = 4P_i\sqrt{s},$$

where $P_i$ is the magnitude of the initial three–momentum and $s = (p_1 + p_2)^2$ is the Mandelstam invariant.

**2. Phase Space Factor** The number of final states is expressed through the invariant phase space element

$$dQ = \frac{d^3p_3}{(2\pi)^3 2E_3}\frac{d^3p_4}{(2\pi)^3 2E_4}(2\pi)^4\delta^{(4)}(p_1 + p_2 - p_3 - p_4).$$

where the first two terms denote number of final states. In the CM frame, this reduces to

$$dQ = \frac{1}{16\pi^2}\frac{P_f}{\sqrt{s}}d\Omega_{CM},$$

where $P_f$ is the magnitude of the final three–momentum.

**Final Expression for $d\sigma$**

Substituting $W_{fi}$, $F$, and $dQ$ into

$$d\sigma = \frac{|\mathcal{M}|^2\, dQ}{F},$$

we obtain

$$d\sigma = \frac{|\mathcal{M}|^2}{F}\frac{1}{16\pi^2}\frac{P_f}{\sqrt{s}}d\Omega_{CM}.$$

Using $F = 4P_i\sqrt{s}$ in the CM frame gives the final result

$$\boxed{\frac{d\sigma}{d\Omega_{CM}} = \frac{1}{64\pi^2 s}\left(\frac{P_f}{P_i}\right)|\mathcal{M}|^2}$$

which is the general two–body differential scattering cross section in the center of mass frame.

This result shows that the observable cross section depends only on Lorentz–invariant quantities: the Mandelstam variable $s$, the invariant amplitude $\mathcal{M}$, and momentum magnitudes defined in the CM frame.

# 3 Relativistic Wave Equations

## 3.1 From Schrödinger Equation to the Klein–Gordon Equation

The need for a relativistic wave equation arises from the incompatibility between the Schrödinger equation and special relativity. In non-relativistic quantum mechanics, the Schrödinger equation is first order in time,

$$i\hbar\frac{\partial}{\partial t}\psi = \hat{H}\psi,$$

with Hamiltonian $\hat{H} = \frac{\hat{p}^2}{2m} + V$. However, this form does not respect Lorentz invariance. To construct a relativistic equation, we begin with the relativistic energy-momentum relation,

$$E^2 = p^2 + m^2,$$

(in natural units $\hbar = c = 1$). Promoting energy and momentum to operators,

$$E \to i\partial_t, \qquad \mathbf{p} \to -i\nabla,$$

we obtain

$$\left(\partial_t^2 - \nabla^2 + m^2\right)\phi = 0,$$

or in covariant notation,

$$(\Box + m^2)\phi = 0,$$

where $\Box = \partial^\mu\partial_\mu$. This is the **Klein–Gordon equation**, the relativistic wave equation for a free spin-0 particle.

## 3.2 Limitations of the Klein–Gordon Equation

Although Lorentz invariant, the Klein–Gordon equation suffers from conceptual difficulties when interpreted as a single-particle wave equation.

**1. Negative Probability Density** The conserved current associated with the Klein–Gordon equation is

$$j^\mu = \frac{i}{2m}\left(\phi^*\partial^\mu\phi - \phi\partial^\mu\phi^*\right).$$

The time component,

$$j^0 = \frac{i}{2m}\left(\phi^*\partial_t\phi - \phi\partial_t\phi^*\right),$$

is not positive definite. Therefore, it cannot be interpreted as a probability density, unlike $|\psi|^2$ in Schrödinger theory.

**2. Negative Energy Solutions** The equation admits both positive and negative energy solutions,

$$E = \pm\sqrt{p^2 + m^2}.$$

In a single-particle interpretation, negative energy states lead to instability. This issue later finds resolution within quantum field theory as antiparticles.

**3. Scalar Nature of the Theory** The Klein–Gordon field transforms as a Lorentz scalar. It therefore describes only spin-0 particles and cannot account for fermions such as electrons, which carry spin-$\frac{1}{2}$.

These limitations motivated the search for a relativistic equation that is first order in time and compatible with spin-$\frac{1}{2}$.

## 3.3 The Dirac Equation

Dirac proposed a linear equation in both time and space derivatives,

$$(i\gamma^\mu \partial_\mu - m)\psi = 0.$$

satisfying

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}.$$

## 3.4 How the Dirac Equation Resolves the Limitations

**1. Positive Definite Probability Density** The conserved current derived from the Dirac equation is

$$j^\mu = \bar{\psi}\gamma^\mu\psi,$$

where $\bar{\psi} = \psi^\dagger\gamma^0$. The time component becomes

$$j^0 = \psi^\dagger\psi,$$

which is manifestly positive definite. Thus, it admits a consistent probability interpretation.

**2. Negative Energy Solutions and Antiparticles** The Dirac equation still contains negative energy solutions, but their interpretation naturally leads to the prediction of antiparticles. In quantum field theory, these solutions correspond to creation operators for antiparticle states, resolving the instability problem present in the Klein–Gordon single-particle interpretation.

**3. Spin-$\frac{1}{2}$ Structure** Unlike the scalar Klein–Gordon field, the Dirac field transforms as a spinor under Lorentz transformations.

## 3.5 Dirac Spinors and Lorentz Transformations

Under a Lorentz transformation $\Lambda$, the Dirac field transforms as

$$\psi'(x') = S(\Lambda)\psi(x),$$

where $S(\Lambda)$ is a $4 \times 4$ matrix representation of the Lorentz group satisfying

$$S^{-1}(\Lambda)\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\nu \gamma^\nu.$$

The Dirac spinor therefore forms a representation of the Lorentz group distinct from scalar or vector representations. Understanding the structure of $S(\Lambda)$, boosts, and rotations in spinor space is essential for constructing relativistically invariant theories of fermions.

I am currently studying the detailed structure of Dirac spinors, their transformation properties under Lorentz transformations, and the explicit construction of spinor solutions in different representations. This forms the next phase of my progress toward a deeper understanding of relativistic quantum theory.

# 4 Introduction to Quantum Field Theory

## 4.1 Quantization of Klein-Gordon Field

**Analogy Between Discrete Quantum Mechanics and Field Theory**

| **Harmonic Oscillator (Quantum Mechanics)** | **Klein–Gordon Field (Quantum Field Theory)** |
|---|---|
| Upon quantization, the introduction of ladder operators $a$ and $a^\dagger$ satisfying $[a, a^\dagger] = 1$ leads to the Hamiltonian $H = \hbar\omega\left(a^\dagger a + \frac{1}{2}\right)$. | Quantization introduces operators $a_k$ and $a_k^\dagger$ satisfying $$[a_k, a_{k'}^\dagger] = \delta^{(3)}(\mathbf{k} - \mathbf{k}').$$ The Hamiltonian becomes a sum over all modes, $$H = \sum_{\mathbf{k}} \hbar\omega_k \, a_{\mathbf{k}}^\dagger a_{\mathbf{k}}$$ |
| The excitation of the system corresponds to increasing the energy level $\lvert n\rangle$, where each action of $a^\dagger$ creates one quantum of energy. | The excitation of a given mode corresponds to the creation of a particle with momentum $\mathbf{k}$. |
| The ground state of the oscillator represents the lowest energy state, and successive applications of the raising operator increase the number of quanta. | The vacuum state $\lvert 0\rangle$ represents the state with no particles, and successive applications of $a_k^\dagger$ build multi-particle states. |

Thus, a quantum field may be understood as an infinite collection of harmonic oscillators, one for each momentum mode, with particles emerging as quantized excitations of the field.

Performing the Legendre transformation,

$$H = \sum_k (p_k \dot{q}_k) - L, \tag{4.1}$$

the Hamiltonian becomes

$$H = \sum_k \left( \frac{1}{2} p_k p_{-k} + \frac{1}{2} \omega_k^2 q_k q_{-k} \right), \tag{4.2}$$

which is a sum of independent harmonic oscillator Hamiltonians.

The Hamiltonian reduces to

$$H = \sum_k \hbar \omega_k \left( a_k^\dagger a_k + \frac{1}{2} \right). \tag{4.3}$$

Redefining the zero-point energy,

$$H = \sum_k \hbar \omega_k a_k^\dagger a_k. \tag{4.4}$$

**Classical Klein–Gordon Field**

We begin with the Lagrangian density of a real scalar field,

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \, \partial^\mu \phi - \frac{1}{2} m^2 \phi^2. \tag{4.5}$$

Applying the Euler–Lagrange equation,

$$\partial_\mu \left( \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0, \tag{4.6}$$

we obtain the Klein–Gordon equation,

$$(\Box + m^2)\phi = 0, \tag{4.7}$$

where $\Box = \partial^\mu \partial_\mu$.

**Fourier Expansion and Mode Equation**

Imposing periodic boundary conditions in a box of volume $V$, the field may be expanded as

$$\phi(t, \mathbf{x}) = \frac{1}{\sqrt{V}} \sum_{\mathbf{k}} q_k(t) e^{i\mathbf{k}\cdot\mathbf{x}}. \tag{4.8}$$

Substituting this expansion into the Klein–Gordon equation,

$$(\partial_t^2 - \nabla^2 + m^2)\phi = 0, \tag{4.9}$$

and using

$$\partial_t^2 \phi = \frac{1}{\sqrt{V}} \sum_k \ddot{q}_k e^{ikx}, \qquad \nabla^2 \phi = -\frac{1}{\sqrt{V}} \sum_k k^2 q_k e^{ikx},$$

we obtain

$$\sum_k \left( \ddot{q}_k + (k^2 + m^2) q_k \right) e^{ikx} = 0. \tag{4.10}$$

Since the plane waves are linearly independent, each mode satisfies

$$\ddot{q}_k + \omega_k^2 q_k = 0, \tag{4.11}$$

where

$$\omega_k^2 = k^2 + m^2. \tag{4.12}$$

Thus, every Fourier mode behaves as a harmonic oscillator.

**Field Operator Representation**

Solving the harmonic oscillator equation yields

$$q_k(t) = \frac{1}{\sqrt{2\hbar\omega_k}} \left( a_k e^{-i\omega_k t} + a_{-k}^\dagger e^{i\omega_k t} \right). \tag{4.13}$$

Substituting back, the field operator becomes

$$\phi(t, \mathbf{x}) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left( a_k e^{-ik\cdot x} + a_k^\dagger e^{ik\cdot x} \right), \tag{4.14}$$

where $k \cdot x = \omega_k t - \mathbf{k} \cdot \mathbf{x}$.

**Canonical Commutation Relation**

The conjugate field momentum is

$$\Pi(t, \mathbf{x}) = \dot{\phi}(t, \mathbf{x}). \tag{4.15}$$

Quantization requires the equal-time commutation relation

$$[\phi(t, \mathbf{x}), \Pi(t, \mathbf{x}')] = i\hbar \delta^3(\mathbf{x} - \mathbf{x}'). \tag{4.16}$$

This condition ensures that the ladder operators satisfy

$$[a_k, a_{k'}^\dagger] = (2\pi)^3 \delta^3(k - k'). \tag{4.17}$$

**Extension to Complex Scalar Fields**

For a complex scalar field, $\phi \neq \phi^\dagger$, two independent sets of oscillators are required. The field expansions become

$$\phi(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left( b_k e^{-ik\cdot x} + a_k^\dagger e^{ik\cdot x} \right), \tag{4.18}$$

$$\phi^\dagger(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left( a_k e^{-ik\cdot x} + b_k^\dagger e^{ik\cdot x} \right). \tag{4.19}$$

Here $a_k^\dagger$ creates particles, while $b_k^\dagger$ creates antiparticles. Thus, complex scalar fields naturally describe charged particles and their corresponding antiparticles.

## 4.2 Interacting Theory

### From Free Theory to Interactions

The central objective of interacting quantum field theory is to express physical quantities in the full interacting theory in terms of the solvable free theory. Specifically, we aim to express the interacting vacuum $|\Omega\rangle$ in terms of the free vacuum $|0\rangle$, and the interacting field $\phi(x)$ in terms of the free field $\phi_0(x)$.

We begin with the free scalar field Lagrangian density,

$$\mathcal{L}_0 = \frac{1}{2}(\partial_\mu \phi_0)(\partial^\mu \phi_0) - \frac{1}{2} m^2 \phi_0^2. \tag{4.20}$$

The corresponding action is

$$S = \int d^4x \left[ \frac{1}{2}(\partial_\mu \phi_0)^2 - \frac{1}{2} m^2 \phi_0^2 \right]. \tag{4.21}$$

Varying the action yields the Klein–Gordon equation,

$$(\Box + m^2)\phi_0 = 0. \tag{4.22}$$

The conjugate momentum is

$$\pi_0(x) = \frac{\partial \mathcal{L}_0}{\partial \dot{\phi}_0} = \dot{\phi}_0(x), \tag{4.23}$$

and the free Hamiltonian becomes

$$H_0 = \int d^3x \left[ \frac{1}{2}\pi_0^2 + \frac{1}{2}(\nabla \phi_0)^2 + \frac{1}{2} m^2 \phi_0^2 \right]. \tag{4.24}$$

This Hamiltonian generates time evolution in the free vacuum $|0\rangle$.

### The Feynman Propagator

The propagation of particles in the free vacuum is encoded in vacuum expectation values. The simplest two-point function is

$$D(x-y) = \langle 0 | \phi_0(x) \phi_0(y) | 0 \rangle. \tag{4.25}$$

The physically relevant object is the time-ordered vacuum expectation value, the Feynman propagator,

$$D_F(x-y) = \langle 0 | T\{\phi_0(x)\phi_0(y)\} | 0 \rangle. \tag{4.26}$$

Using step functions, this becomes

$$D_F(x-y) = \theta(x^0 - y^0)\langle 0|\phi_0(x)\phi_0(y)|0\rangle + \theta(y^0 - x^0)\langle 0|\phi_0(y)\phi_0(x)|0\rangle. \tag{4.27}$$

After performing the Fourier transform and implementing the $i\epsilon$ prescription to regulate pole structure, one obtains the momentum-space representation,

$$D_F(x-y) = \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} e^{-ik(x-y)}. \tag{4.28}$$

This propagator serves as the fundamental building block of perturbation theory.

**Introducing Interactions: $\phi^4$ Theory**

To describe scattering processes, we introduce a non-linear interaction term. The interacting Lagrangian is defined as

$$\mathcal{L} = \mathcal{L}_0 - V(\phi_0), \tag{4.29}$$

with interaction potential

$$V(\phi_0) = \frac{\lambda}{4!}\phi_0^4. \tag{4.30}$$

The full Lagrangian becomes

$$\mathcal{L} = \frac{1}{2}(\partial_\mu \phi_0)^2 - \frac{1}{2}m^2\phi_0^2 - \frac{\lambda}{4!}\phi_0^4. \tag{4.31}$$

Applying the Euler–Lagrange equation now yields the non-linear equation of motion,

$$(\Box + m^2)\phi_0 + \frac{\lambda}{3!}\phi_0^3 = 0. \tag{4.32}$$

The cubic term breaks linear superposition and allows different field modes to exchange energy, leading to scattering phenomena. The coupling constant $\lambda$ controls the strength of interaction. In four spacetime dimensions, $\lambda$ is dimensionless, ensuring renormalizability of the theory.

**The Total Hamiltonian**

Since the interaction term does not involve time derivatives, the conjugate momentum remains $\pi_0 = \dot{\phi}_0$. The Hamiltonian density becomes

$$\mathcal{H} = \mathcal{H}_0 + \frac{\lambda}{4!}\phi_0^4. \tag{4.33}$$

Thus the full Hamiltonian is

$$H = H_0 + H_{\text{int}}, \tag{4.34}$$

where

$$H_{\text{int}} = \int d^3x \frac{\lambda}{4!}\phi_0^4. \tag{4.35}$$

**The Interaction Picture and Time Evolution**

The interacting field evolves according to

$$\phi(t, \mathbf{x}) = e^{iHt}\phi(\mathbf{x}, 0)e^{-iHt}. \tag{4.36}$$

The free field evolves as

$$\phi_0(t, \mathbf{x}) = e^{iH_0t}\phi_0(\mathbf{x}, 0)e^{-iH_0t}. \tag{4.37}$$

Defining the unitary operator

$$U(t) = e^{iH_0t}e^{-iHt}, \tag{4.38}$$

we obtain the fundamental relation

$$\phi(t, \mathbf{x}) = U^{-1}(t)\phi_0(t, \mathbf{x})U(t). \tag{4.39}$$

Taking the time derivative of $U(t)$ yields

$$i\frac{d}{dt}U(t) = H_I(t)U(t), \tag{4.40}$$

where

$$H_I(t) = e^{iH_0t}H_{\text{int}}e^{-iH_0t}. \tag{4.41}$$

The formal solution is the time-ordered exponential,

$$U(t) = T\left\{\exp\left(-i\int_0^t dt'\, H_I(t')\right)\right\}. \tag{4.42}$$

**The Dyson Series and Perturbation Theory**

Expanding the exponential generates the Dyson series,

$$\begin{aligned} U(t) = 1 - i\int_0^t dt_1 H_I(t_1) \\ + \frac{(-i)^2}{2!}\int_0^t dt_1 dt_2\, T\{H_I(t_1)H_I(t_2)\} + \cdots . \end{aligned} \tag{4.43}$$

Each term corresponds to a definite number of interaction vertices and directly maps

onto Feynman diagrams. For small coupling $\lambda$, the first few terms provide an accurate approximation.

**The S-Matrix**

The scattering matrix is defined by evolving from the far past to the far future,

$$S = T\left\{\exp\left(-i\int_{-\infty}^{\infty} dt\, H_I(t)\right)\right\}. \tag{4.44}$$

The matrix elements $\langle f|S|i\rangle$ give probability amplitudes for physical scattering processes.

**The Interacting Vacuum**

To obtain the interacting vacuum, we evolve the free vacuum under the full Hamiltonian and project onto the ground state,

$$|\Omega\rangle = \lim_{t\to\infty(1-i\epsilon)} \frac{e^{-iHt}|0\rangle}{e^{-iE_0 t}\langle\Omega|0\rangle}. \tag{4.45}$$

Using this construction, the interacting correlation functions can be expressed entirely in terms of free fields:

$$\langle\Omega|T\{\phi(x_1)\cdots\phi(x_n)\}|\Omega\rangle = \frac{\langle 0|T\{\phi_0(x_1)\cdots\phi_0(x_n)e^{-i\int d^4x H_{\text{int}}(\phi_0)}\}|0\rangle}{\langle 0|T\{e^{-i\int d^4x H_{\text{int}}(\phi_0)}\}|0\rangle}. \tag{4.46}$$

The denominator cancels vacuum bubble diagrams, ensuring proper normalization. Thus, all interacting physics is reduced to free-field calculations weighted by interaction insertions.

# 5 Introduction to Quantum Information Theory

## 5.1 Different inequalities predicted by LHVT

**Local Hidden Variable Theories**

Local Hidden Variable Theories (LHVT) were proposed to restore a classical intuition to quantum mechanics. Following the Einstein–Podolsky–Rosen (EPR) argument, it was suggested that quantum mechanics might be incomplete, and that measurement outcomes are determined by pre-existing hidden variables $\lambda$ that are not included in the standard quantum formalism.

Two central assumptions define LHVT:

- **Realism**: Measurement outcomes are predetermined by hidden variables.
- **Locality**: A measurement performed at one location cannot instantaneously influence outcomes at a spacelike separated location.

Under these assumptions, joint correlations between distant measurements can be written as

$$E(a,b) = \int d\lambda\, \rho(\lambda) A(a,\lambda) B(b,\lambda),$$

where $A, B = \pm 1$ are predetermined outcomes and $\rho(\lambda)$ is a probability distribution.

Bell-type inequalities place constraints on such correlations. Quantum mechanics predicts violations of these inequalities for entangled states.

**1. Bell's Inequality** For two observers measuring along directions $a, b, c$, a Local Hidden Variable Theory implies

$$\boxed{|E(a,b) - E(a,c)| \leq \frac{\hbar^2}{4} + E(b,c)}$$

where $E(a,b)$ denotes the expectation value of the product of spin observables.

**Violation using the Singlet State** Consider the spin singlet Bell state,

$$|\psi^-\rangle = \frac{1}{\sqrt{2}} \left( |\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle \right).$$

Quantum mechanics predicts

$\langle\psi^-|(\hat{S}_a \otimes \hat{S}_b)|\psi^-\rangle = -\frac{\hbar^2}{4}\, a \cdot b.$

Choose measurement directions separated by 60°:

$\theta_{ab} = \theta_{bc} = 60°, \qquad \theta_{ac} = 120°.$

Then, $E(a,b) = E(b,c) = -\frac{\hbar^2}{4} \cos 60° = -\frac{\hbar^2}{8},$

$E(a,c) = -\frac{\hbar^2}{4} \cos 120° = +\frac{\hbar^2}{8}.$

Substituting into Bell's inequality,

$|E(a,b) - E(a,c)| = \left| -\frac{\hbar^2}{8} - \frac{\hbar^2}{8} \right| = \frac{\hbar^2}{4},$

while

$\frac{\hbar^2}{4} + E(b,c) = \frac{\hbar^2}{4} - \frac{\hbar^2}{8} = \frac{\hbar^2}{8}.$

Hence, $\frac{\hbar^2}{4} > \frac{\hbar^2}{8},$

which violates Bell's inequality.

**2. CHSH Inequality** The CHSH inequality is a more experimentally useful form. For two measurement settings per observer, LHVT predicts

$$\boxed{|S| := |\langle a_1 b_1\rangle + \langle a_1 b_2\rangle + \langle a_2 b_1\rangle - \langle a_2 b_2\rangle| \leq 2}$$

**Quantum Violation** Using the singlet state with

$$\theta_{a_1 b_1} = 0°, \quad \theta_{a_1 b_2} = 45°, \quad \theta_{a_2 b_1} = 45°, \quad \theta_{a_2 b_2} = 90°,$$

and the quantum prediction

$\langle a_i b_j \rangle = -\cos\theta_{a_i b_j}$,

we obtain

$S = -\cos 0^\circ - \cos 45^\circ - \cos 45^\circ + \cos 90^\circ$.

Evaluating,

$S = -1 - \frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2} + 0 = -\left(1 + \sqrt{2}\right)$.

Thus,

$|S| = 1 + \sqrt{2} \approx 2.414 > 2$ The maximal quantum violation is

$|S|_{\max} = 2\sqrt{2}$.

**3. Leggett–Garg Inequality**

Unlike Bell inequalities, the Leggett–Garg inequality tests *macrorealism* and *non-invasive measurability* in time rather than spatial separation.

For an observable $Q(t) = \pm 1$ measured at three times $t_1, t_2, t_3$, define temporal correlations

$C_{ij} = \langle Q(t_i) Q(t_j) \rangle$.

$$\boxed{K = C_{12} + C_{23} - C_{13} \leq 1.}$$

**Quantum Violation Example** Consider a two-level system evolving under

$$H = \frac{\omega}{2}\sigma_x.$$

If the system is initially prepared in an eigenstate of $\sigma_z$, quantum mechanics predicts

$$C_{ij} = \cos\left[\omega(t_i - t_j)\right].$$

Choose equal time intervals $\tau$:

$C_{12} = C_{23} = \cos(\omega\tau), \quad C_{13} = \cos(2\omega\tau)$.

Then

$K = 2\cos(\omega\tau) - \cos(2\omega\tau)$.

Using $\cos(2x) = 2\cos^2 x - 1$,

$K = 2\cos x - (2\cos^2 x - 1) = 1 + 2\cos x(1 - \cos x)$.

For suitable choice of $x$, for example $x = \pi/3$, $K = \frac{3}{2} > 1$,

violating the Leggett–Garg inequality.

These violations demonstrate that no theory satisfying locality and realism (or macrorealism in the temporal case) can reproduce all quantum predictions.

## 5.2 Measures of Entanglement

### Entanglement in Pure States

**Definition of a Pure State** A bipartite quantum system composed of subsystems $A$ and $B$ is described by a pure state if its density operator satisfies

$$\rho = |\psi\rangle\langle\psi|, \qquad \text{with } \mathrm{Tr}(\rho^2) = 1.$$

Such a state can be written as a normalized vector in the tensor product Hilbert space $\mathcal{H}_A \otimes \mathcal{H}_B$.

**Separable and Entangled Pure States** A pure state $|\psi\rangle$ is called *separable* if it can be written as a product state,

$$|\psi\rangle = |\phi_A\rangle \otimes |\phi_B\rangle.$$

If no such decomposition exists, the state is *entangled*.

Using the Schmidt decomposition, any bipartite pure state can be written as

$$|\psi\rangle = \sum_i \sqrt{\lambda_i}\, |i_A\rangle|i_B\rangle,$$

where $|i_A\rangle|i_B\rangle$ are orthonormal bases, $\lambda_i \geq 0$ are Schmidt coefficients and $\sum_i \lambda_i = 1$. The state is separable if and only if exactly one Schmidt coefficient is nonzero.

**Measure of Entanglement for Pure States** The standard measure of entanglement for a pure state is the *von Neumann entropy* of the reduced density matrix.

The reduced density matrix for subsystem $A$ is

$$\rho_A = \mathrm{Tr}_B(|\psi\rangle\langle\psi|).$$

The entanglement entropy is defined as

$$E(|\psi\rangle) = S(\rho_A) = -\mathrm{Tr}(\rho_A \ln \rho_A).$$

For a Schmidt decomposition, this becomes

$$E(|\psi\rangle) = -\sum_i \lambda_i \ln \lambda_i.$$

**Example: Bell State** Consider the maximally entangled Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

The reduced density matrix is $\rho_A = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

Thus, $E = -\left(\frac{1}{2}\ln_2\frac{1}{2} + \frac{1}{2}\ln_2\frac{1}{2}\right) = ln2$.

Hence, Bell states possess one ebit of entanglement.

**Entanglement in Mixed States**

**Definition of a Mixed State** A mixed state is described by a density operator

$$\rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|,$$

where $0 < p_i < 1$, $\sum_i p_i = 1$, and $\mathrm{Tr}(\rho^2) < 1$.

**Separable and Entangled Mixed States** A mixed state is separable if it can be written as

$$\rho = \sum_i p_i \, \rho_A^{(i)} \otimes \rho_B^{(i)}.$$

If no such decomposition exists, the state is entangled.

For mixed states, entanglement cannot be quantified simply by the entropy of a reduced density matrix, and more refined measures are required.

# References


[1] D. J. Griffiths, *Introduction to Elementary Particles*, 2nd Edition, Wiley-VCH, 2008.

[2] F. Halzen and A. D. Martin, *Quarks and Leptons: An Introductory Course in Modern Particle Physics*, Wiley, 1984.

[3] F. Schwabl, *Advanced Quantum Mechanics*, 4th Edition, Springer, 2008.

[4] M. D. Schwartz, *Quantum Field Theory and the Standard Model*, Cambridge University Press, 2014.

[5] F. Mandl and G. Shaw, *Quantum Field Theory*, 2nd Edition, Wiley, 2010.